Closed-loop discovery of NIPS membranes

Robotic synthesis / compression-derived properties / vision-guided experimental decisions

01 Define the experiment

Fixed: polymer, base solvent, cosolvent and NIPS-bath solvent identity / wt%
Seed: filtered history or initial recipe
8 controls: polymer / additive wt%, mix / bath temperature, casting speed, air-exposure time, N_2, immersion time

run_loop.py · operator configuration

02 Fabricate autonomously

Opentrons OT-2 + xArm + bath chiller
Dose stocks → heat / mix → blade cast
Wait in ambient air or dry nitrogen
Immerse in non-solvent bath → retrieve
Record ambient and coupon T / RH
Robot receives recipe via POST /run

protocol-multithreaded.py · robot PC

03 Measure & archive

4 zero + 4 sample compression curves
Pre-test and post-test photographs
LVDT position check before testing
Batch completion: POST /server/process
Archive 8 CSVs + 2 images + params.json
One condition folder per synthesis run

run_loop.py · data/raw/<condition>/

Two enabled analysis branches[1]

04 Extract properties & quality

MECHANICS: chronological zero pairing; compliance subtraction → stress–strain
GAM + regional fits; thickness / fit checks
E, yield, creep, slopes, fixed-stress strain
Pore Fraction ≔ fitted changepoint strain
VISION: pre-test photo → quality report

curve_segmentation.py · membrane_imaging.py

05 Build experimental memory

reps.csv: individual measurement results
agg.csv: mean ± SD; pre / post-discard
llm.csv: promoted post-discard outcomes
Attach fit failures + visual quality report
Join current campaign + warm-start data
Retain each run's material / bath context

llm_context.py · property_context.py

06 Choose & validate next run

LLM: improve modulus + pore fraction
Balance Pareto gains with exploration
Prompt requests independent validation
Return 8-control JSON → schema check
Apply additive lock + stock feasibility
Undiluted stock: mixing set to 25 °C

activeLearning_29.py · bounds · run_loop.py

Next feasible recipe

OFFLINE EVALUATION

Fit / quality review → HTML logs | Repeatability → matched synthesis groups → between-run SD / CV

Exact: polymer wt%, N_2, bath-solvent wt%. Tolerances: wait 10 s; bath 3 °C; cast 3 mm/s; RH 5 pp.

[1] Branches dispatch sequentially. Checkpoints support recovery; ambiguous robot submission requires operator review.